Project Report
Genetic Disorder Risk Prioritization Platform

Executive Summary & Verification Report

This platform addresses the prioritization of high-risk genetic disorder patient records using machine learning. Rather than using an arbitrary threshold (e.g., 50%), we evaluated dynamic probability thresholds to optimize patient outreach.

Data Quality & Preprocessing Strategy

* **Identifiers Dropped**: Non-informative unique IDs (like `Patient Id`) were stripped out to prevent target leakage.

* **Imputation**: Categorical values were imputed with the mode, and numerical attributes (like `Patient Age`) were imputed with column medians.

* **Label Encoding**: Categorical variables were label encoded to match the numerical structure required by the XGBoost algorithm.

Model Performance & Threshold Analysis

Below is the summary of the evaluated thresholds for classifying critical status:

* **Threshold 0.3**: Highest Recall (47% detection rate of risk instances), making this threshold optimal for conservative safety programs looking to capture as many potential cases as possible.

* **Threshold 0.7**: Highest Precision (maximizing accuracy on flagged patients), appropriate when clinical staff resources are severely constrained and false positives must be kept to a minimum.

***Selected Threshold for Platform Default**: **0.4 to 0.5** yields a stable compromise between capture rate (Recall) and false alarm rates.